UNIVERSITY OF CAPE TOWN
IYUNIVESITHI YASEKAPA • UNIVERSITEIT VAN KAAPSTAD

Statistical Sciences

STA5092Z – Exploratory Data Analysis
Assignment 2 & 3

Late submissions will be penalised at 5% per day (pro rata)

NOTE:
This assignment consists of two sections. Make sure to attempt both sections. Broadly speaking your goals are:

1. Data wrangling and plotting of data: **Section 1** (Assignment 2) **[50 marks]**
2. Creating a Shiny app for exploring data: **Section 2** (Assignment 3) **[50 marks]**

Be sure to read and follow the instructions given in each section carefully.

This exercise is based on data on recorded earthquakes around the world from 1965 until March 2023. The data come from two separate sources:

The file `Earthquakes 1965- 2016.csv` was obtained from this Kaggle page. It con tains measurements on 21 variables of recorded earthquakes (or earthquake-like phenom ena) ranging from 2 January 1965 until 30 December 2016. 2. The file query.csv was generated from the USGS website, and contains measurements on 22 features from 1 December 2016 until 17 March 2023.

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SECTION 1: Assignment 2 [50 marks]

Instructions:
• Write up your work in the form of a short report (maximum 10 pages) in whatever typesetting software you prefer (e.g. MSWord, LATEX, RMarkdown, etc.)
• The report should contain an introduction to the problem, but the majority of the document should be about your results.
• Your code should be submitted as well, either embedded in the document if a .Rmd file, or as a separate .R or .Rmd file. The key thing is that I need to be able to run your code and reproduce your results. Use comments!
• The code itself should NOT be displayed in the final typeset document (use "echo = FALSE" for .Rmd files) and NOT pasted as an appendix in your report. • Remember to attach a plagiarism declaration to your report.

In this exercise we will explore data on recorded earthquakes around the world from 1965 until March 2023. The data come from two separate sources, requiring some wrangling into a single data frame. We will then investigate some temporal and spatial aspects of the earthquakes.

1. Data wrangling
These files must be read directly into R without editing; no wrangling may be implemented in Excel. The first task is to merge them into one dataframe/tibble. You will notice that the features in the two datasets only partially overlap, and that the naming convention differs between them. Only use common variables, i.e. similar information contained in both datasets. Briefly document your process and reasoning in creating a single dataframe.

Note:
- There must only be one date-time variable (note that all times are UTC).
- Provide a summary of the types of events and then filter the observations to only include "Earthquake" type events.
- Add a categorical variable "Scale", which classifies the earthquakes according to their magnitude. Classify and name the categories according to the "Description" column in the table on this page.

2. Initial exploration
Before investigating the temporal and spatial aspects of the data, explore and describe the features. Things we might be interested in include:

- The distributions of the features. Decide which features to investigate and provide the appropriate plots/measurements.

- The largest earthquakes, according to magnitude. Choose a number of them and present and discuss in an appropriate way.

3. Temporal investigation

We will now investigate the frequency and severity of earthquakes over time. For this section you have freedom to choose what to focus on and how to present it, in lieu of step-by-step instructions. Some of the questions you might like to answer are:

- How do the frequency of earthquakes in each category compare to estimated frequencies?
- Do we observe any patterns over time? Is this a definite function of time, or perhaps an artefact of the data?

4. Spatial exploration

This task is again left mostly open-ended, with the goal being to provide a summary of the spatial distribution of the earthquakes. You may incorporate the temporal nature of the data in any way you see fit.

Note that all figures may be static, i.e. no interactivity (since it is a pdf report) and no animation is expected, although you are welcome to include some if you feel you need to! After providing the global picture, focus on three specific regions, with the corresponding questions merely acting as guidelines/suggestions:

SECTION 2: Assignment 3 [50 marks]

Instructions:

Your project consists of the following items, with the % contribution to the final project mark:

1. Your R Shiny app– 80%. This should be an R script called STDNUM001.R (replace with your student ID) made up of a user interface object (ui) and a server function (server).
2. Documentation– in a file named `STDNUM001.pdf`– explaining how to use the app and providing a short worked example– 20%. This is a "read me" or "help file" for the user, showcasing the app's functionalities.

Please note the following:

- The app must read in all the .csv files from a directory called data. This directory must resides in the same directory as `STDNUM001.R`. This directory must also contain your data.
- All files must be read in as provided.
- All code in the app should be clearly commented.
- Please make sure that your code/app works by testing it out first on another machine before submitting it.
- The code itself should not be displayed in your help documentation (the runner is not interested in the back end).
- You may of course publish your app to your shinyapps.io dashboard and include the link in your documentation file, but this is not a requirement and will not earn extra marks. Remember to upload a plagiarism declaration

In this exercise you will create a shiny app to explore the earthquake data. **Use your merged and cleaned data set**. This assignment is open-ended. Use your imagination and insights gained from Section 1 (Assignment 1) to decide what kind of app you want to create, and which aspects of the data you want to be able to explore with your app.